Milestone Report: Learning to Select Density Functionals for GSCDB Reaction Energies

CS285 Project — RL Functional Selector

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Abstract

We study the problem of choosing among a fixed set of density-functional approximations (DFAs) for predicting GSCDB reaction energies from molecular geometry alone. A neural policy is trained with supervised warm-start followed by batched REINFORCE, using pooled features from reactant and product `.xyz` files. On a held-out test split, the greedy policy achieves substantially lower mean absolute error (MAE) than choosing a functional uniformly at random, but remains well above an oracle that always picks the best DFA per reaction using tabulated energies. Training curves show rapid improvement during supervised pretraining and a plateau in classification-style metrics during the reinforcement-learning phase.

1 Question and motivation

Accurate reaction energies depend strongly on the choice of electronic-structure method. The Gold-Standard Chemical Database (GSCDB) provides reference reaction energies and tabulated values for many popular DFAs. The motivating question is:

> *Given only the geometries of the species involved in a reaction (reactants and products), can we predict which DFA will yield a reaction energy closest to the reference?*

This is framed as a *discrete decision problem*: the action space is the set of DFAs present in `Reaction_Energies.csv`. Success is measured by (i) how often the model's top choice coincides with a lowest-error functional, and (ii) the MAE of the reaction energy for the predicted functional versus the reference.

2 Method

2.1 Data and labels

- **Reactions:** Entries from `Info/DatasetEval.csv` are aligned with `Analysis/Reaction_Energies.csv`. Stoichiometry strings are parsed so that negative stoichiometric coefficients correspond to *reactants* and positive coefficients to *products*, consistent with the database's energy-difference convention.
- **Geometry input:** For each species, features are extracted from `GSCDB/xyz_files/<id>.xyz`: atom counts, coarse geometry statistics (e.g. radius of gyration, extrema of interatomic distances), and normalized element fractions. Reactant-side and product-side species are pooled (coefficient-weighted means) and concatenated with simple counts to form a fixed-length state vector per reaction.

- **Supervision signal:** For each reaction, the "best" functional index is the argmin of $|E_j - E_{\text{ref}}|$ over DFAs j. Training rewards in REINFORCE use $r = -|E_{\hat{j}} - E_{\text{ref}}|$ for the sampled action $\hat{j}$.

2.2 Policy and training

- **Architecture:** A multi-layer perceptron with ReLU hidden layers outputs logits over DFAs; a softmax defines a stochastic policy $\pi(a \mid s)$.
- **Supervised warm-start:** Cross-entropy minimization toward the argmin-error label (batched SGD) provides a low-variance initial signal.
- **REINFORCE:** Batched policy-gradient updates with per-batch centered advantages reduce variance relative to single-sample updates. An entropy bonus encourages exploration. A moving average of batch-mean reward is logged as a training baseline (not a performance metric on the test set).
- **Evaluation split:** Reactions are shuffled; approximately 85% train / 15% test (typical run: ∼7,121 train / ∼1,256 test reactions when the full GSCDB subset with complete XYZ is used).

2.3 Metrics

- **Test greedy accuracy:** Fraction of test reactions for which the greedy argmax functional achieves the minimum $|E - E_{\text{ref}}|$ (ties count as correct).
- **Test MAE (greedy):** Mean over test reactions of $|E_{\hat{a}} - E_{\text{ref}}|$ where $\hat{a}$ is the greedy action (Hartree).
- **Oracle MAE:** Mean over test reactions of $\min_j |E_j - E_{\text{ref}}|$ — the best achievable MAE if one always selected an optimal DFA from the tabulated list (uses energies at test time; not realizable from geometry alone).
- **Uniform-random MAE:** Monte Carlo estimate of the MAE when j is drawn uniformly over DFAs (sanity baseline).
- **Probability on best:** Average total policy probability placed on all functionals tied for minimum error (smoother than greedy accuracy).

3 Results

3.1 Representative numbers (one full training run)

Table 1 summarizes metrics consistent with logged training output: supervised warm-start improves greedy accuracy and MAE; after extended REINFORCE, greedy test accuracy can plateau while MAE remains between oracle and random baselines.

The model clearly **beats random selection** on MAE but is **far from oracle**: simple geometry descriptors do not fully determine which functional wins across diverse GSCDB sets.

Table 1: Illustrative test-set metrics (order of magnitude from a representative run with ∼1,256 test reactions and 16 DFAs).

Quantity	Value	Comment
Test greedy accuracy	~ 0.16	$\approx 16\%$ top-1 hits vs. $\sim 1/16 \approx 6\%$ random
Test MAE (greedy)	~ 0.76 Ha	~ 480 kcal/mol
Oracle MAE	~ 0.19 Ha	Lower bound among listed DFAs
Uniform-random MAE	~ 1.7 Ha	Weak baseline

3.2 Training dynamics (figures)

Figure 1 shows four panels over cumulative updates (warmup indices $1 \ldots W$, then REINFORCE $W{+}1, \ldots$). **Top-left:** test and train greedy accuracy rise during supervised warmup; the vertical line marks the switch to REINFORCE. **Top-right:** test MAE in Hartree with oracle (green) and random (orange) reference lines — the learned curve should lie between them when the policy is informative. **Bottom-left:** probability mass on truly best functionals. **Bottom-right:** REINFORCE reward baseline (training diagnostic).

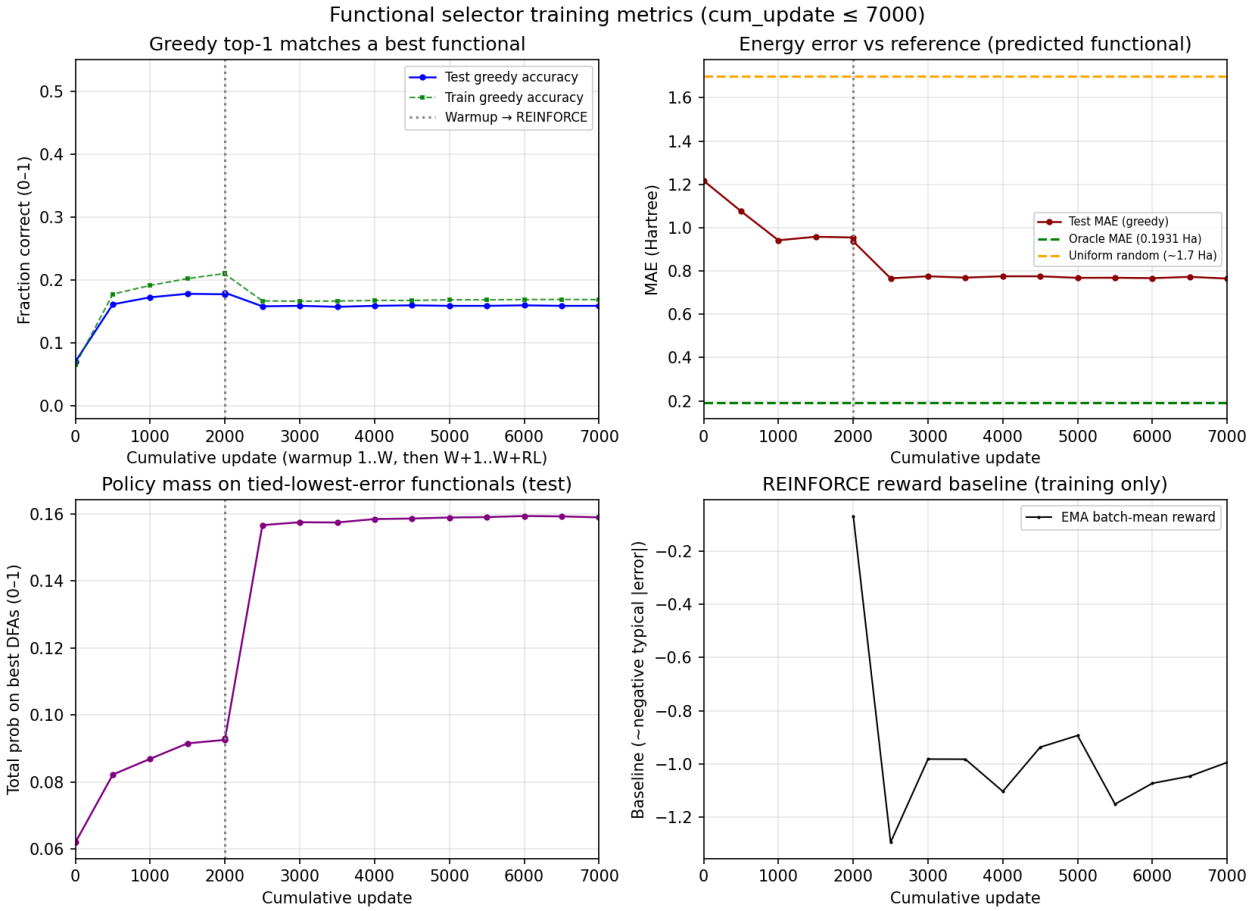

Figure 1: Training metrics for the first 7000 cumulative logged updates (warmup + early REINFORCE).

Figure 2 plots the same test MAE in kcal/mol for interpretability (1 Hartree $\approx$ 627.5 kcal/mol).

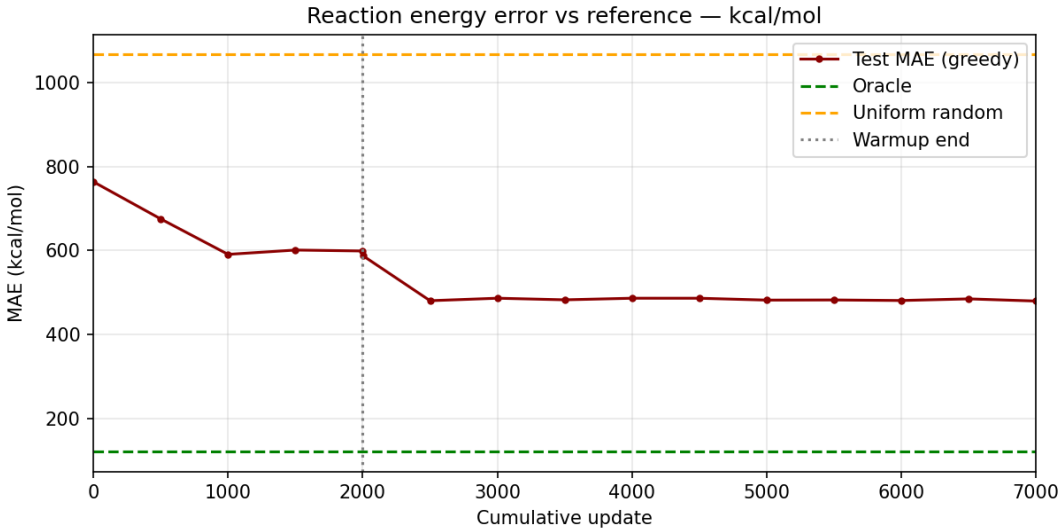

Figure 2: Test MAE versus reference in kcal/mol (first 7000 cumulative updates).

4 Discussion

Greedy accuracy is a *hard* metric: small logit changes rarely flip the argmax, so curves can appear flat even while MAE or probability-on-best move. REINFORCE optimizes expected sampled reward, not greedy test accuracy, which can explain post-warmup regressions or plateaus in top-1 accuracy. The gap to oracle indicates that stronger representations (e.g. dataset identity, richer descriptors, or graph-based models) would likely be needed for large gains.

5 Next steps

- Extend the state with dataset identifiers or stoichiometry metadata.
- Train primarily with supervised or soft-label objectives aligned with MAE or ranking.
- Report confidence intervals over multiple random seeds and splits.

Reproducibility. Train: `python train_reaction_reinforce.py --gscdb-root ../../GSCDB --out-dir ../models/reaction_reinforce`. Plots: `python plot_training_curves.py --history ../models/reaction_reinforce/training_history.json --max-cum-update 7000`. Compile this document from the `RL_Functional_Selector` directory so figure paths resolve.